MAT01B1 Final: Exam Day Sheet

Thursday 22 October 2026, test from 12:30 to 14:30. Written the night before from your three 2022 papers and tonight's session.

1. Where you stand

Paper	Mark
2022 Exam A	about 47 / 70 (67%)
2022 Exam C	about 45 / 70 (64%)
2022 Exam B	about 50 / 70 (72%)
Semester Test 2	66%, with about 75% average going into the final

Your goal is 80%. You do not have a knowledge problem. Almost every lost mark was one line: a wrong radius, a dropped $\frac{1}{x}$, a sign, a missing "there exists". Each of those is caught by a check you can do slowly. That is the plan for tomorrow.

2. The five habits that cost you marks

Habit 1: Volumes of revolution (lost marks in A, B and C)

- Draw the axis and **one strip** before any formula.
- Say the radius out loud in words: "outer radius is ..., inner radius is ...".
- Axis $y = -2$ below the region: radius is $y + 2$, so $2 + \cos x$ and $2 + \sin x$, not $2 - \cos x$.
- Test the value at one point. A volume can never be negative.
- If two methods are possible, do both. If they disagree, there is a slip.
- Common slips: $x \cdot \sqrt{x} = x^{3/2}$, and solving $y = \sqrt{27}\sqrt{x}$ for x gives $x = \frac{y^2}{27}$.

Habit 2: One-line formula and arithmetic slips

- **Write the formula from memory before using it.** $\displaystyle\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \arctan \frac{x}{a}$, so with $a = 2$ it is $\frac{1}{2} \arctan \frac{x}{2}$.
- **Arithmetic on its own line.** $-2 - 40 = -42$, not -38.
- **Know your values.** $\csc \frac{\pi}{6} = 2$, $\frac{\sin^2 \theta}{\cos^2 \theta} = \tan^2 \theta$.
- **Copy the whole expression every line.** $x \arctan\left(\frac{1}{x}\right)$ kept its $\frac{1}{x}$ on line 1 and lost it on line 3.
- **After every integral, differentiate once.** $\int 6y^2 dy = 2y^3$, not $3y^3$. And $\int x e^{x^2} dx = \frac{1}{2} e^{x^2}$, using $u = x^2$.

Habit 3: Read the restriction

- MVT needs differentiable on the open interval. $\sqrt[3]{x}$ has $f'(x) = \frac{1}{3x^{2/3}}$, which does not exist at 0, so the MVT cannot be applied on $[-1, 1]$.
- $x = \cos t$ only runs from -1 to 1, so $y = 1 - x^2$ is a **portion** of a parabola.
- Domain first: square roots need the inside ≥ 0, and a denominator cannot be 0.

Habit 4: Finish what the question asks

- "Find the local maximum **values**": write $f(0) = 0$ and $f(4) = 8$, not just "at 0 and 4".
- Asymptotes: say all three, including "no horizontal asymptote".
- A graph with a vertical asymptote needs a branch on **both sides** of it.
- Improper integral: end with "convergent" or "divergent" and the value.
- Every theorem statement ends with "there exists c..." and every hypothesis is a mark.
- Simplify the final answer, and label every part, such as 13(b) and 5(e).

Habit 5: Time

- You ran over in Exam C. Circle the question that stalls you and move on. If you blank, write the setup or the first line of the theorem, because setup lines earn marks.

3. Exam time plan (12:30 to 14:30)

This assumes a paper like the 2022 ones: 70 marks in 120 minutes. The rule is **about 1.5 minutes per mark**.

Time	What to do
12:30 to 12:35	Skim the whole paper. Star the questions you can do fast.
12:35 to 12:47	Question 1, multiple choice (10 marks). Work it out yourself, then match to an option. Never choose "none of the above" first.
12:47 to 14:15	Written questions, about 60 marks. Do the starred ones first.
14:15 to 14:30	Check: differentiate integral answers, recheck volume radii, theorem statements, multiple choice.

Checkpoints: by 13:00 about 19 marks done, by 13:30 about 39, by 14:00 about 60, by 14:15 everything attempted.

Minutes per question: 2 marks is about 3 minutes, 3 marks about 4.5, 4 marks about 6, 6 marks about 9. If the paper is a different size, use $1.5 \times \text{marks}$.

If you blank: write the first line, circle it, move on. Come back with the time you saved.

4. Thursday morning plan

Shift the times to your wake-up, but keep the order. No new material after 11:00.

Time	Block
07:00	Wake, breakfast, water.
07:45 to 08:45	**Block 1, choosing the method** (Section 5). Do the three practice setups, then redo Exam B Question 7 and Question 8 on a blank page.
08:45 to 09:00	Break. Walk and water.
09:00 to 10:00	**Block 2, proofs** (Section 6). Read once, then write the MVT via Rolle, the MVT for integrals, Fermat and Rolle on paper.
10:00 to 10:15	Break.
10:15 to 11:00	**Block 3, slip fixes.** Redo on a blank page: Exam B 2(a), 3, 6, 9, 11(b), 13(b), then Question 4.
11:00 to 11:15	Read Sections 2 and 3 once, then close everything.
11:15	Stop. Light lunch, pack, travel. Be at the venue by 12:00.

Bring: student card, two pens, pencil, ruler, calculator if allowed, water.

5. Choosing the method

Keyword table

The question says	You use
"area of the region between curves"	$\int (\text{top} - \text{bottom})\, dx$
"average value"	$f_{ave} = \dfrac{1}{b-a}\displaystyle\int_a^b f\, dx$
"length of the curve"	$L = \displaystyle\int ds$
"area of the surface obtained by rotating"	$S = \displaystyle\int 2\pi(\text{radius})\, ds$
"volume of the solid obtained by rotating"	washers or shells

where $ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$ for $y = f(x)$, $ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\,dy$ for $x = g(y)$, and $ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$ for a parametric curve.

Washers or shells

1. Sketch the region and draw one strip.
2. Strip **perpendicular** to the axis (it cuts across it): **washers**, $V = \pi \int (R^2 - r^2)$.
3. Strip **parallel** to the axis: **shells**, $V = 2\pi \int (\text{radius})(\text{height})$.
4. Horizontal axis: washers use dx, shells use dy. Vertical axis: washers use dy, shells use dx.
5. If the question names the method, follow it.

The radius: swap the letters

Axis	Radius
the y-axis	x
the x-axis	y
the line $x = a$	$\lvert x - a \rvert$
the line $y = k$	$\lvert y - k \rvert$

The axis is named by one letter, and the radius is the **other** letter. Picture test: spin the point $(3, 1)$ about the y-axis and you get a circle of radius 3.

Shells in four steps

1. **Strip direction:** vertical axis gives vertical strips (dx). Horizontal axis gives horizontal strips (dy).
2. **Radius** from the table above.
3. **Height** is the strip length: far curve minus near curve, in the strip's variable. Test one number to see which curve is on top (or on the right).
4. **Limits:** where the strip starts and stops, from the intersection points, in the same variable.

Surface area in four steps

1. Name the **curve** and the axis.
2. **Radius** = distance from a point on the curve to the axis (table above).
3. **ds** in the variable the curve is given in, or the variable the question names. The radius must be written in the **same variable**.

4. Limits in that variable. $S = \int 2\pi(\text{radius})\,ds$.

Worked examples

- **A.** Region between $y = x$ and $y = x^2$ about the y-axis: $V = 2\pi \int_0^1 x(x - x^2)\,dx = \dfrac{\pi}{6}$.
- **B.** Same region about $x = 2$: radius is $2 - x$, so $V = 2\pi \int_0^1 (2 - x)(x - x^2)\,dx = \dfrac{\pi}{2}$.
- **C.** Region under $y = x^2$ for $0 \le x \le 1$ about the x-axis, using shells: $x = \sqrt{y}$, so $V = 2\pi \int_0^1 y\,(1 - \sqrt{y})\,dy = \dfrac{\pi}{5}$.
- **D.** Curve $y = x^2$, $0 \le x \le 1$, about the y-axis: $S = 2\pi \int_0^1 x\sqrt{1 + 4x^2}\,dx$.
- **E.** Same curve about the x-axis: $S = 2\pi \int_0^1 x^2\sqrt{1 + 4x^2}\,dx$.
- **Exam B Q10:** $y = \sqrt[3]{x}$, $1 \le y \le 2$, about the y-axis in y: $x = y^3$, $S = 2\pi \int_1^2 y^3\sqrt{1 + 9y^4}\,dy$.
- **Exam B Q8:** $y = x^2$ and $y = \sqrt{27}\sqrt{x}$ about the y-axis. Meeting at $x = 3$. Shells: $2\pi \int_0^3 x\left(\sqrt{27}\sqrt{x} - x^2\right)dx = \dfrac{243\pi}{10}$. Washers: $\pi \int_0^9 \left(y - \dfrac{y^4}{729}\right)dy = \dfrac{243\pi}{10}$.

Practice setups (answers below, cover them first)

1. Region between $y = \sqrt{x}$ and $y = x^2$ rotated about $x = -1$, using shells.
2. The same region rotated about $y = -1$, using shells.
3. The curve $y = x^3$, $0 \le x \le 1$, rotated about the x-axis: surface area.

Answers. 1. $V = 2\pi \int_0^1 (x + 1)\left(\sqrt{x} - x^2\right)dx$. 2. Horizontal axis, so dy. The curves in terms of y are $x = y^2$ and $x = \sqrt{y}$, with $\sqrt{y}$ on the right. $V = 2\pi \int_0^1 (y + 1)\left(\sqrt{y} - y^2\right)dy$. 3. $S = 2\pi \int_0^1 x^3\sqrt{1 + 9x^4}\,dx$.

6. Proofs (all of them)

The hypotheses are marks. In every statement, write "there exists".

6.1 Mean Value Theorem, via Rolle

Statement. If f is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ with $f'(c) = \dfrac{f(b) - f(a)}{b - a}$.

Proof. Define

$$h(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a)$$

Picture: h is the vertical gap between the graph and the straight line through the two end points.

- h is continuous on $[a, b]$ and differentiable on (a, b).
- $h(a) = f(a) - f(a) - 0 = 0$ and $h(b) = f(b) - f(a) - \big(f(b) - f(a)\big) = 0$.
- Equal ends, so Rolle gives $c \in (a, b)$ with $h'(c) = 0$.
- $h'(c) = f'(c) - \dfrac{f(b) - f(a)}{b - a} = 0$, which gives the result.

6.2 MVT for integrals (EVT and IVT proof)

Statement. If f is continuous on $[a, b]$, then there exists $c \in [a, b]$ with $f(c) = f_{ave} = \dfrac{1}{b - a} \displaystyle\int_a^b f(x)\, dx$.

Proof. - By the Extreme Value Theorem, there are $m, M \in [a, b]$ where f has its absolute minimum $f(m)$ and maximum $f(M)$. - So $f(m) \le f(x) \le f(M)$ for all $x \in [a, b]$. - Integrate: $f(m)(b - a) \le \displaystyle\int_a^b f(x)\, dx \le f(M)(b - a)$. The ends are constants, so $\int_a^b f(m)\, dx = f(m)(b - a)$. - Divide by $b - a$: $f(m) \le f_{ave} \le f(M)$. - f is continuous, so by the Intermediate Value Theorem it takes every value between $f(m)$ and $f(M)$. So there exists $c \in [a, b]$ with $f(c) = f_{ave}$.

6.3 Fermat's Theorem

Statement. If f has a local maximum or minimum at c, and $f'(c)$ exists, then $f'(c) = 0$.

Proof. Assume a local maximum at c. Then $f'(c) = \lim\limits_{h \to 0} \dfrac{f(c + h) - f(c)}{h}$ and, for small h, $f(c + h) \le f(c)$.

- $h > 0$: the quotient is ≤ 0, so $f'(c) = \lim_{h \to 0^+} \le 0$.
- $h < 0$: the quotient is ≥ 0, so $f'(c) = \lim_{h \to 0^-} \ge 0$.
- $f'(c)$ exists, so both one-sided limits are equal. So $0 \le f'(c) \le 0$, and $f'(c) = 0$.

A local minimum works the same way. The converse is false: $f(x) = x^3$ at 0 has $f'(0) = 0$ and no extremum.

6.4 Rolle's Theorem

Statement. If f is continuous on $[a, b]$, differentiable on (a, b) and $f(a) = f(b)$, then there exists $c \in (a, b)$ with $f'(c) = 0$.

Proof. - **Case I:** f is constant. Then $f' = 0$ everywhere, so any c works. - **Case II:** $f(x) > f(a)$ for some $x \in (a, b)$. By the EVT, f has a maximum on $[a, b]$. It is bigger than $f(a) = f(b)$, so it is not at an end point. It is at some $c \in (a, b)$, a local maximum, so Fermat gives $f'(c) = 0$. - **Case III:** $f(x) < f(a)$ for some x. Same argument with the minimum.

6.5 Exactly one root (Exam C 3b: $x^3 + x - 3 = 0$)

Let $f(x) = x^3 + x - 3$, continuous and differentiable everywhere.

- **At least one root:** $f(0) = -3 < 0$ and $f(2) = 7 > 0$, so by the Intermediate Value Theorem there is a root in $(0, 2)$.
- **At most one root:** suppose two roots $r_1 < r_2$. Then $f(r_1) = f(r_2) = 0$, and Rolle gives $c \in (r_1, r_2)$ with $f'(c) = 0$. But $f'(x) = 3x^2 + 1 \geq 1 > 0$ for all x. Contradiction.
- So there is exactly one root.

6.6 Polar slope

Statement. For $r = f(\theta)$, with $r' = \dfrac{dr}{d\theta}$:

$$\frac{dy}{dx} = \frac{r' \sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$$

Derivation. Use θ as the parameter: $x = r\cos\theta$, $y = r\sin\theta$. By the product rule, $\dfrac{dx}{d\theta} = r'\cos\theta - r\sin\theta$ and $\dfrac{dy}{d\theta} = r'\sin\theta + r\cos\theta$. Then $\dfrac{dy}{dx} = \dfrac{dy/d\theta}{dx/d\theta}$. This needs $\dfrac{dx}{d\theta} \neq 0$.

6.7 Binomial sigma form

Statement. For natural n:

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k, \qquad \binom{n}{k} = \frac{n!}{(n-k)!\,k!}$$

Pattern for "show that". Put in a and b, then combine the powers of x in each term. Example: $\left(x + \frac{1}{x}\right)^5 = \sum_{k=0}^{5} \dfrac{5!}{k!\,(5-k)!}\, x^{5-2k}$. The coefficient of x^3: $5 - 2k = 3$, so $k = 1$ and the coefficient is 5.

Finding a term. Add the powers of x in a^{n-k} and b^k, set equal to the target, solve for k. Include the signs and numbers in a and b. Exam B Q15: $\left(x^3 - \frac{2}{x^2}\right)^{12}$, term in x: $36 - 5k = 1$ gives $k = 7$, coefficient $\binom{12}{7}(-2)^7 = 792 \cdot (-128) = -101376$.

6.8 Circumference as an improper integral

Statement. For $x^2 + y^2 = 1$, $L = 2\displaystyle\int_{-1}^{1} \frac{dx}{\sqrt{1-x^2}}$.

Derivation. The upper semicircle is $y = \sqrt{1-x^2}$, so $\dfrac{dy}{dx} = \dfrac{-x}{\sqrt{1-x^2}}$ and

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{x^2}{1-x^2} = \frac{1}{1-x^2}, \qquad ds = \frac{dx}{\sqrt{1-x^2}}$$

The circumference is twice the semicircle. The integrand is undefined at $x = \pm 1$, so the integral is improper.

6.9 Lower priority, short forms

- **Theorem 5.** If $f' = 0$ on (a, b), then f is constant. *Proof:* take any $x_1 < x_2$, and the MVT gives $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0$.

- **Corollary 7.** If $f' = g'$, then $f = g + c$. *Proof:* $F = f - g$ has $F' = 0$, so F is constant. This is why every antiderivative carries $+C$.
- **Increasing test.** If $f' > 0$ on an interval, f is increasing. *Proof:* for $x_1 < x_2$, the MVT gives $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) > 0$. For $f' < 0$, the same with a negative product.
- **Shell volume.** One shell is outer cylinder minus inner cylinder: $\pi r_2^2 h - \pi r_1^2 h = 2\pi \frac{r_1 + r_2}{2} h (r_2 - r_1)$, which is circumference × height × thickness. Cut $[a, b]$ into strips, each shell has radius $\bar{x}_i$, height $f(\bar{x}_i)$ and thickness Δx, so $V = \lim \sum 2\pi \bar{x}_i f(\bar{x}_i) \Delta x = \displaystyle\int_a^b 2\pi x f(x)\, dx$.
- **Arc length.** Join points P_i on the curve by straight segments: $|P_{i-1}P_i| = \sqrt{(\Delta x)^2 + (\Delta y_i)^2}$. By the MVT, $\Delta y_i = f'(x_i^*)\Delta x$, so each piece is $\sqrt{1 + [f'(x_i^*)]^2}\, \Delta x$. Sum and limit give $L = \displaystyle\int_a^b \sqrt{1 + [f'(x)]^2}\, dx$.
- **Surface area.** Each segment rotated gives a band of area $2\pi \cdot (\text{average radius}) \cdot |P_{i-1}P_i|$. With the same arc length step, the sum becomes $S = \displaystyle\int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2}\, dx$.
- **Integrating factor.** For $y' + Py = Q$, want $I(y' + Py) = (Iy)'$. Expanding gives $IP = I'$, which is separable: $\frac{dI}{I} = P\, dx$, so $I = e^{\int P\, dx}$. Then $(Iy)' = IQ$.

7. What you are supposed to know

Theorem statements

1. **Fermat:** local max or min at c and $f'(c)$ exists, then $f'(c) = 0$.
2. **Rolle:** continuous on $[a, b]$, differentiable on (a, b), $f(a) = f(b)$, then there exists $c \in (a, b)$ with $f'(c) = 0$.
3. **MVT:** continuous on $[a, b]$, differentiable on (a, b), then there exists $c \in (a, b)$ with $f'(c) = \frac{f(b) - f(a)}{b - a}$.
4. **MVT for integrals:** continuous on $[a, b]$, then there exists $c \in [a, b]$ with $f(c) = f_{ave}$.
5. **Intermediate Value Theorem:** continuous on $[a, b]$, N between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ with $f(c) = N$.
6. **Extreme Value Theorem:** continuous on a closed interval attains a maximum and a minimum.

Integration

- By parts: $\int u\, dv = uv - \int v\, du$.
- Partial fractions: linear factor gets a constant, a repeated factor gets one term per power, an irreducible quadratic gets $Ax + B$.
- Trig substitution: $\sqrt{a^2 - x^2}$ uses $x = a \sin\theta$, $\sqrt{a^2 + x^2}$ uses $x = a \tan\theta$, $\sqrt{x^2 - a^2}$ uses $x = a \sec\theta$.

- Improper integral: write the limit, evaluate, say convergent or divergent.
- $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\arctan\frac{x}{a}$, $\int \sec x\,dx = \ln|\sec x + \tan x|$, $\int \csc x\,dx = \ln|\csc x - \cot x|$ (or $-\ln|\csc x + \cot x|$).
- Identities: $1 + \tan^2 x = \sec^2 x$, $1 + \cot^2 x = \csc^2 x$, $\sin 2\theta = 2\sin\theta\cos\theta$, $\cos 2\theta = \cos^2\theta - \sin^2\theta$.

Differential equations

- **Separable:** $\int \frac{dy}{g(y)} = \int f(x)\,dx$, then use the initial condition for C.
- **Linear:** $y' + Py = Q$, $I = e^{\int P\,dx}$, $(Iy)' = IQ$. Divide by the coefficient of y' first.

Parametric and polar

- $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$, $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}$.
- Polar: $x = r\cos\theta$, $y = r\sin\theta$, $r^2 = x^2 + y^2$. For a polar equation with $\sin\theta$ or $\cos\theta$, multiply by r.

Curve sketching and related rates

- **Sketch order:** domain, intercepts, asymptotes (vertical, horizontal, slant), f' sign table (increase, decrease, local values), f'' sign table (concavity, inflection), then the sketch.
- **Related rates:** draw and label, write the equation, differentiate with respect to t, and **substitute numbers only after differentiating**.

Matrices and linear systems

- AB exists when columns of A equal rows of B, and has the size rows of A by columns of B.
- Trace is the sum of the diagonal (square matrix only). The transpose swaps rows and columns.
- Gauss-Jordan: reduce to leading ones with zeros above and below. A free variable is a column with no leading one: give it a parameter.
- A row of zeros with a non-zero number on the right means no solution.
- Check the answer in the original equations.

8. Last lines

You finished full papers tonight with correct methods on integration, matrices, differential equations, the binomial question and the MVT proof. The marks you lost are the kind a calm, slow recheck recovers. Draw the axis. Write the formula. Differentiate your answer. Write "there exists". Move on when stuck.

Good luck.